Revised Project Proposal: Fault-Tolerant Control Under Complete Rotor Loss via Physics-Informed Residual Koopman MPC

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1. Summary of Revisions

Following your feedback on our earlier proposal, we went back and re-audited the literature in detail. You were right: both "Koopman-based control for quadrotors" and "Koopman-based control for actuator faults" already exist in the 2024–2026 literature — including recent work applying residual Koopman modeling to quadrotors directly. We are grateful for the correction, since it forced us to find a more precise and defensible research question rather than a generic application of a known method.

The revised proposal below identifies a narrower, more specific gap and states our novelty claim carefully — as something we did not find in an extensive search, rather than as an absolute claim.

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2. The Revised Research Question

> **Can a Koopman/EDMDc residual model, built around the analytically known spinning-equilibrium dynamics of a completely failed quadrotor, provide a better long-horizon predictive model for MPC than either the analytical failure model alone or a generic full-state Koopman model?**

What is genuinely different from existing work, and why we believe it is defensible:

- Existing Koopman fault-tolerant control (e.g., Lee et al., 2024) addresses **partial loss-of-effectiveness** faults, where all four rotors remain active and the vehicle retains full attitude authority.
- Existing residual Koopman work on quadrotors (Todde et al., 2026) and quadrupeds (Narayanan & Vaidya, 2026) addresses **nominal operation under model/parameter uncertainty**, not actuator loss.
- **Complete rotor loss is structurally different from both**: one column of the actuator-mixing map vanishes entirely, yaw authority is lost, and the vehicle is forced into a qualitatively different regime — a continuous spinning equilibrium — rather than a degraded version of normal hover.

We could not find residual Koopman modeling applied specifically to this complete-loss, spinning-equilibrium regime, across searches spanning direct term combinations, citation trails from the Mueller & D'Andrea paper, and adjacent "reduced attitude" terminology. We state this as the result of our search, not as a claim that it does not exist anywhere.

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3. The Architecture

1. Complete rotor failure
2. Spinning equilibrium (Mueller & D'Andrea)
3. Learn residual with EDMDc
4. MPC

- A generic Koopman-MPC approach asks a data-driven model to learn the entire post-failure dynamics from scratch.
- In our own preliminary testing (Section 5 below), this produced a numerically unstable learned model.
- Instead, we provide the dominant post-failure physics analytically, via Mueller & D'Andrea's spinning-equilibrium model.
- We then ask the learned model to represent only the mismatch:

$$e_k = x_k^{\text{true}} - x_k^{\text{nominal}}$$

We then build three controllers on the same simulator and compare them directly:

1. **MPC on the analytical nominal model alone** (no learning)
2. **MPC on a full-state Koopman model** (our original approach — retained as a baseline precisely because we found it to be unstable, which motivates the residual formulation)
3. **MPC on the residual Koopman model** (proposed approach)

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4. Why Mueller–D'Andrea Specifically, Not an Arbitrary Nominal Model

This is the central technical justification for the approach, not a citation of convenience.

Complete rotor loss is not "normal flight plus a disturbance." After a rotor is completely lost, the vehicle permanently loses independent control of yaw, and it can no longer hover still — the only valid steady state remaining is a continuous spin about a fixed body axis, with

translational motion achieved by tilting that spin axis. This is a genuinely different equilibrium, not a perturbed version of the previous one.

An arbitrary nominal model would defeat the purpose of residual learning. If we instead used a generic nominal model unaware of the failure — e.g., an ordinary hover template — it would keep predicting a vehicle sitting still and level, while the real vehicle is continuously spinning. The "residual" would then have to contain the entire spinning motion: a large, persistent signal, not a small correction. This does not simplify the learning problem; it relabels it. The benefit of residual learning depends entirely on the nominal model already capturing the *dominant* behavior.

Mueller & D'Andrea's model is dominant-behavior-complete for this specific regime. It is derived specifically for complete rotor loss and already provides, in closed form: the correct spin axis and spin rate, the correct steady-state thrust levels, and the correct relationship between axis tilt and translation. What remains for a residual to capture is genuinely second-order — blade flapping and induced-drag effects at high spin rate, inertia measurement error, motor lag, and similar effects not captured by their idealized derivation. This is plausibly a small, well-behaved residual, which is the regime EDMDc is best suited to.

Two further practical advantages. The model is linear and time-invariant once expressed as deviations from the equilibrium, making it cheap to compute and well understood — unlike attempting to correct around a constantly moving, otherwise-undefined target. It is also not a guess: it is experimentally validated (real quadrotor flight, one- and two-rotor loss) and analytically proven controllable, so the physics half of the architecture is trustworthy before any learning is introduced.

Why we expect this to help, stated as a hypothesis, not a result. The same pattern — a physics backbone with a learned correction for the remaining mismatch — has shown benefits in adjacent domains: soft robot arms, ground vehicles, and quadruped locomotion, consistently producing smaller and better-conditioned models than learning the full dynamics from data. There is a physically grounded reason to expect this to transfer to the complete-rotor-loss regime. Testing whether it does is the actual content of the project, not an assumption we are making going in.

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5. Preliminary Evidence Motivating This Direction

Before arriving at this formulation, we built a proof-of-concept fitting a full-state EDMDc model to simulated (undamaged) quadrotor flight data. Diagnostic testing revealed that although the model was accurate one step ahead (comparable training and held-out error), the learned linear operator had 10 of 22 eigenvalues with magnitude greater than 1 — meaning errors compound exponentially rather than merely accumulate over a rollout, and the model underperformed a

trivial "assume nothing changes" baseline beyond roughly half a second of open-loop prediction. This is a documented failure mode of full-state EDMD in the recent literature (e.g., Narayanan & Vaidya, 2026, who report a related SE(3)-Koopman baseline becoming unstable within seconds in simulation, motivating their own residual formulation). This finding is what motivated us to move from a full-state to a residual formulation, rather than treating it as background reading alone.

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8. Key References

1. Mueller, M.W. & D'Andrea, R. (2014). *Stability and Control of a Quadrocopter Despite the Complete Loss of One, Two, or Three Propellers.* ICRA 2014.
2. Nan, F., Sun, S., Foehn, P., & Scaramuzza, D. (2022). *Nonlinear MPC for Quadrotor Fault-Tolerant Control.* IEEE RA-L.
3. Lee et al. (2024). *Koopman Fault-Tolerant Model Predictive Control for Multirotor with Actuator Fault.*
4. Lee et al. (2024). *Data-Driven Fault Detection and Isolation for Multirotor System Using Koopman Operator.* J. Intelligent & Robotic Systems.
5. Narayanan & Vaidya (2026). *RK-MPC: Residual Koopman Model Predictive Control for Quadruped Locomotion in Offroad Environments.* arXiv:2604.04221.
6. Todde, Martini, Rizzo, & Valavanis (2026). *Residual Koopman-Based Model Predictive Control of Quadrotors.* ICUAS 2026.
7. Bruder, D., Bombara, D., & Wood, R.J. (2025). *A Koopman-based residual modeling approach for the control of a soft robot arm.* IJRR.
8. Fu et al. (2025). *Residual Koopman Model Predictive Control for Enhanced Vehicle Dynamics with Small On-Track Data Input.*